

DigitalTwin.ai — Round 2 Plan of Action

What Round 2 asks, what we already have, what is missing, and the order to build it in. Team HipHipHooray — Sagar Sahu, Priyansh Goyal.

1 · WHAT ROUND 2 REQUIRES

Three deliverables, per the problem-statement document:

DELIVERABLE	WHAT IT MEANS FOR US
Detailed Business Proposal	Problem framing, solution design, target users, business case and impact, a phased roadmap, and key risks with mitigations. Much of this exists in our 21-page design document — it needs a business layer added, not a rewrite.
Working Prototype	A functional demonstration of the core mechanism. Explicitly <i>not</i> required to be production-grade or use real enterprise data — a proof-of-concept on illustrative data is "expected and encouraged".
Pitch Presentation	Presenting proposal and prototype together.

2 · GAP ANALYSIS — ROUND 2 ASKS VS WHAT WE HOLD

ROUND 2 REQUIREMENT	STATUS	ACTION
Uneven sensor coverage; legacy + modern mix	HAVE	Pipeline 3 is built around exactly this. It is our strongest differentiator.
Defect surfaces at a later inspection point; downstream units carry it	PARTIAL	Detection is built; genealogy containment is designed but not built . Build it.
No modification of live PLCs / line control	HAVE	Already an explicit design boundary: the twin advises, never writes.
Multi-causal, intermittent root causes	PARTIAL	Four failure modes separable on three features — classifier not fitted . Build it.
Three stakeholder views — floor supervisor, plant manager, leadership	GAP	We have one implied view. This is a named requirement; build all three.
Predictive claims validated against outcomes; false alarms erode trust	HAVE	Held-out calibration already does this. Surface it in the prototype.
Scaling across lines/plants of different vintage	PARTIAL	Layouts L1–L4 exist for exactly this transfer test — never run . Run it.
30–50 stations across body, paint, final assembly	GAP	Our plant is 20 stations, single segment. Extend to ~40 with three segments.
Business case / ROI / phased roadmap	GAP	We have one cited figure (\$2M/hour downtime). Needs a real case.

THE REASSURING READ

Four of nine are already covered, and two of the gaps (three views, 30–50 stations) are presentation and configuration work rather than new science. The genuinely new build is the

LIVE LOOP

— which we were going to need anyway.

3 · BUILD ORDER

Phases are ordered by dependency. Each has a gate — a concrete pass condition, not a feeling.

Phase 0 — Clear the ground · ½ day

- Fix or delete the global `speed_scale` path; correct the `plant.py` docstring that contradicts its own test.
- Measure the paired-comparison noise floor across ~50 seeds and publish the number.

Gate: 11/11 tests green; noise floor documented.

Phase 1 — Station record · 2–3 days

- `src/twin/record.py` : one row per station per interval, every field carrying **value + provenance** (measured / inferred / predicted) + **confidence**.
- Enforced in code: a field missing any of the three is not emitted.
- Confidence calibrated, not asserted — validate by switching off sensors we have and comparing against known truth.

Gate: a calibration curve — when it says 90%, it is right ~90% of the time (± 10 points).

Phase 2 — Alert contract · 2 days

- An alert may only be raised carrying all five: ranked candidate *and margin*, evidence, persistence estimate, recommended action, expected cost of not acting.
- An alert that cannot state its evidence is **suppressed, not downgraded**.
- Persistence filter ~10 minutes — below that the constraint moves before anyone arrives.

Gate: replay a shift; every alert carries all five fields; the rate is actionable.

Phase 3 — Close the loop · 3–4 days · ESSENTIAL

- `src/twin/loop.py` : ingest window → update → detect → rank → emit → wait. Same code path for replay and live.
- Replay driver over the event log at arbitrary speed — this is also what makes the system auditable after an incident.
- **Twin drift metric:** model-predicted vs observed throughput over a trailing window, with a re-fit trigger that logs the re-fit.

Gate: a full shift replayed at 60× producing a live record stream, with drift visibly triggering a re-fit at a changeover.

Phase 4 — The three stakeholder views · 3 days · NEW FOR R2

- **Floor supervisor:** real-time — what is constraining now, what to do, how confident.
- **Plant manager:** weekly trends — where constraints recur, which buffers threaten, defect warning lead times.
- **Leadership:** the investment case — cars recovered, sensor spend recommended and its payback.

Gate: three views from the *same* underlying record stream, no separate pipelines.

Phase 5 — Scale and close the measured gaps · 3 days

- Extend the plant to ~**40 stations across three segments** (body / paint / final) to match the brief's reference parameters.
- Run the **layout-transfer test** on L1–L4 — built for this, never executed.
- Bias dark-station placement toward the constraint in ~40% of runs so localisation becomes scoreable (today only 5 of 142 blocks are positive).
- Fit the failure-mode classifier on *ratios*, not our generator's sigma values; score it on the Wiener firewall set never used in development.

Gate: a transfer number across layouts; a calibratable dark threshold; a classifier scored on the firewall set.

Phase 6 — Genealogy + the joint decision · 3 days

- **Genealogy containment:** "station 22 has been drifting since 10:14; these 38 vehicles were built on it since." This directly answers Round 2's late-inspection complexity.
- **Stop-or-continue:** weight by $P(\text{constraint})$, never a hard indicator; integrate a *rising* hazard.

Gate: the same drifting tool getting opposite correct answers depending on flow state.

Phase 7 — Proposal, demo and pitch · 4 days

- Business proposal: problem framing, design, users, business case, roadmap, risks + mitigations.
- Demo: switch off sensors we actually have, estimate anyway, reveal the truth alongside. Nothing else proves the method as directly.
- The observability figure loses its ILLUSTRATIVE label because it is finally measured.

Gate: a five-minute walkthrough runnable end to end without touching a script.

4 · SEQUENCE AND EFFORT

PHASE	EFFORT	PRIORITY
0 · Clear the ground	½ d	Blocking
1 · Station record	2–3 d	Core
2 · Alert contract	2 d	Core
3 · Close the loop	3–4 d	Essential
4 · Three stakeholder views	3 d	Required by R2
5 · Scale + measured gaps	3 d	High
6 · Genealogy + joint decision	3 d	Medium
7 · Proposal, demo, pitch	4 d	Ship

Roughly **21 working days** end to end. Working in parallel on separate modules, with gate checks run jointly, that compresses considerably — phases 4 and 5 are largely independent of 1–3.

Suggested split

- **Sagar** — phases 1–3 (record, alerts, loop). This is the spine and must not be split across two people.
- **Priyansh** — phase 0, then 5 (scale to 40 stations, layout transfer, dark placement, classifier). Independent of the loop until phase 4.
- **Together** — phases 4, 6 and 7.

5 · RISKS AND MITIGATIONS

THE NOISE FLOOR MAY MOVE A PUBLISHED NUMBER

If the floor lands near 5 cars, the healthy-run figures sit inside it and must be restated as "no detectable gain".

Mitigation: measure it in phase 0, before it is quoted anywhere. The faulted-run result carries the argument on its own, so this costs us nothing to be honest about.

DARK LOCALISATION MAY STAY UNSCOREABLE

Biased placement raises the positive rate, but the constraint and dark block may still rarely coincide. *Mitigation, decided in advance:* ship

EXONERATION

as the capability — clearing a block is the common and operationally useful case — and report localisation as designed-not-validated.

SCOPE CREEP AWAY FROM THE LOOP

Phases 5 and 6 are more fun than phase 3 and will pull attention. *Mitigation:* a working loop with two pipelines beats three pipelines that only run in batch, because the loop is the claim a judge will actually probe. If time runs short, cut 6 first, then 5 — never 3.

FALSE ALARMS ERODE FLOOR TRUST

Named explicitly in the Round 2 brief. *Mitigation:* the persistence filter and the suppress-without-evidence rule already exist in the design; make them visible in the demo rather than leaving them implicit.

6 · WHAT WE WILL NOT BUILD

Each was considered and cut for a stated reason — saying so is stronger than silence: 3D geometry and CAD visuals (no predictive value); robot kinematics and weld/paint physics (collapse into distributions); write-back to line control (a regulated change no plant grants a prototype); a graph neural network over topology (no evidence it beats gradient boosting on a near-serial line); and **drift-based overtake risk** — built, tested, failed at 5.9% against 70–100% stated confidence. It stays cut, and we keep reporting the failure.